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SUPERSEDING
MIL-T-8429A(USAF)
29 May 1957

MILITARY SPECIFICATION

TESTER, LEAKAGE, DEMAND OXYGEN REGULATOR, TYPE MH-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable tester to determine the leakage in demand oxygen regulators, designated type MH-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

*	QQ-A-250/8F	-	Aluminum Alloy 5052, Plate and Sheet.
	RR-C-271	-	Chains and Attachments, Welded and Weldless.
*	TT-L-32A	-	Lacquer, Cellulose Nitrate, Gloss for Aircraft Use.
*	TT-P-1757	-	Primer Coating, Zinc Chromate, Low-Moisture Sensitivity.

MILITARY

	MIL-R-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes.
*	DOD-D-1000	-	Drawing, Engineering and Associated List.
*	MIL-C-52211	-	Components and Assemblies for Industrial Gas Production Storage and Transportation Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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STANDARDS

FEDERAL

FED-STD-595	-	Color

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-143	-	Standards and Specifications, Order of.
* MIL-STD-831	-	Test Reports, Preparation of.
* MIL-STD-889	-	Dissimilar Metals.
* MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.

DRAWINGS

AIR FORCE

55B3510	-	Connector, Demand Mask to Regulator Tube Internal.
55A3512	-	Gasket, Demand Mask to Regulator Tube Internal.
57B3623	-	Ring-Retaining, Demand Mask to Regulator Tube Internal.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Component parts. The tester shall consist of a case, a flow-measuring gage, a pressure gage, a suitable orifice, a rubber bulb and pressure reservoir, a pressure bleed valve necessary tubing and adapters for connecting the unit to the outlet of the regulator, and an operating instruction card.

3.4 Materials.

3.4.1 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected against corrosion during normal service life. Nonferrous materials shall be used in all parts of the tester except where ferrous materials are essential and do not come into contact with the oxygen. Nonmagnetic materials shall be used for all parts of the tester, except where magnetic materials are essential.

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3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Elastomer components. The synthetic rubber parts used in the tester shall not be more than eight quarters old from the time of cure date to the time of delivery to the Government. The cure date (quarter and year) shall be stamped on the elastomer component, providing there is sufficient area. The figures shall be at least 3/16 inch high. Cure dates and age limitations shall not apply to silicone. All elastomers shall be free from agents which may cause objectionable or intolerable odors.

3.4.3 Protective treatment. When materials are used in the construction of the tester that are subject to corrosion in oxygen, salt spray, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.5 Design. The complete tester shall conform substantially to Figure 1. The components shall be enclosed in a metal case, as shown in Figure 1. The case shall be provided with a carrying handle.

3.6 Performance.

3.6.1 General. With the adapter inserted into the fitting on the mask to regulator tubing, and with pressure applied by the rubber bulb and pressure reservoir, the flow measuring gage shall indicate the leakage through the regulator and tubing in liters per minute.

3.6.2 Flow measuring gage. The flow-measuring gage shall be calibrated with orifice attached. It shall be calibrated in the range of 0 to 2.0 liters per minute with a pressure of 17 inches of water indicated on the pressure gage. Calibration shall be accomplished at room temperature with the use of a master flow meter or other flow measuring device which will be measured with the accuracy and the range of the gage to be calibrated. The flow measuring gage shall be a sensitive differential pressure-type indicator capable of withstanding a maximum of 3 inches of mercury pressure. The dial shall be calibrated in hundredths of a liter from 0 to 2.0 liter per minute flow. The gage shall operate satisfactorily within the scale range of any differential pressure produced by the orifice. The gage mechanism shall be provided with suitable stops to prevent gage damage from overflow or reversed flow. The gage's pressure taps shall have orifices that are suitable to dampen gage response in the case of sudden flow changes. The flow meter shall be marked: "Leakage Flow, Liters Per Minute." The flow-measuring gage shall have plus or minus 1 percent accuracy at full scale. The gage shall have a diameter of approximately 3 1/8 inches. The calibration shall be from 0 to 2.0 liters per minute, to be calibrated starting at the top over at least 270 degrees of arc. The dial face shall be as shown in Figure 2.

3.6.3 Pressure gage. The pressure gage accuracy shall be within plus or minus 3 percent at full scale. The gage shall be capable of withstanding a maximum pressure of 3 inches of mercury. The gage mechanism shall be provided with suitable stops to prevent over deflection and damage to the mechanism.

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This gage shall be connected to the downstream side of the orifice, and its pressure connections shall have a suitable orifice to dampen gage responses and to assist in protecting the gage in the event of sudden pressure changes. The gage shall be approximately 3 1/8 inches in diameter. The calibration shall be from 0° to 30 inches of water in units of 1 inch. The pressure gage shall be marked: "Pressure - Inches of Water." The dial face shall be as shown in Figure 2.

3.6.4 Leakage. Leakage through the tester shall be determined with the adapter plugged and with 20 inches of water pressure applied in the system. The leakage shall not be such that the rate of pressure drop is greater than 0.50 inch of water per minute nor shall the flow measuring gage deviate from zero (in accordance with section 5 of 3.10, operating instructions).

3.6.5 Overload. The tester shall be capable of withstanding an internal pressure of 3 inches of mercury for a period of 2 minutes without damage.

3.6.6 High temperature operation. The tester shall be capable of satisfactory operation at a temperature of plus 160 degrees Fahrenheit.

3.6.7 Low temperature operation. The tester shall be capable of satisfactory operation at a temperature of minus 65 degrees Fahrenheit.

3.6.8 Shock. When tested in accordance with 4.5.7, the gages shall show no evidence of leakage, permanent deformation, or loosening of parts, and shall retain their serviceability. A maximum cumulative deviation from absolute accuracy of plus or minus 5 percent of all values above 25 percent of full scale, and for zero shift, as a result of these tests is permissible. This permitted deviation is an overall deviation and is not in addition to the initial accuracy tolerance.

3.7 Details of components.

3.7.1 Removable filter. The tester shall have a conveniently replaceable filter of 100-gage mesh screen or equivalent. This filter shall adequately protect all working parts from dust and other foreign particles, and shall not impair flow measurements.

3.7.2 Orifice. A suitable orifice shall be incorporated in the system which is capable of producing the necessary pressure differentials for deflecting the flow measuring gage within the specified range, when the downstream pressure varies from 0 to 20 inches of water.

3.7.3 Pressure bulb and reservoir. Pressure shall be applied to the oxygen regulator through the tester by means of a hand operated rubber bulb fabricated of synthetic rubber conforming to MIL-R-6855, class II, equipped with a pressure reservoir. The rubber bulb and pressure reservoir shall be equal to and interchangeable with Arthur H. Thomas Co., Philadelphia, PA, Part Number 8784. A pressure-bleed valve shall be located between the bulb and reservoir to allow deflation of the reservoir prior to storage.

3.7.4 Tubing. The tubing used between the pressure reservoir and the tester and for connecting the tester to the regulator outlet shall be 3/16 inch inside diameter light-wall tubing conforming to MIL-R-6855, class II. Each piece of tubing shall be approximately 30 inches long.

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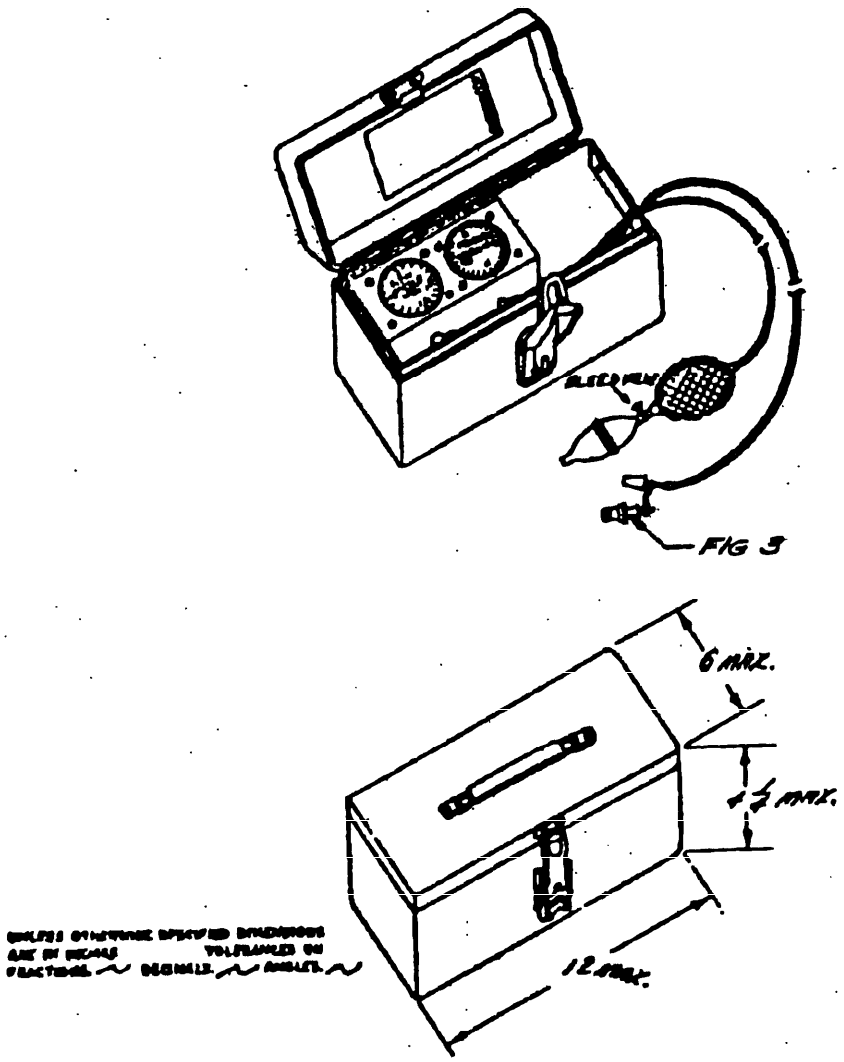


FIGURE 1. Tester, leakage, demand oxygen regulator, type MH-2

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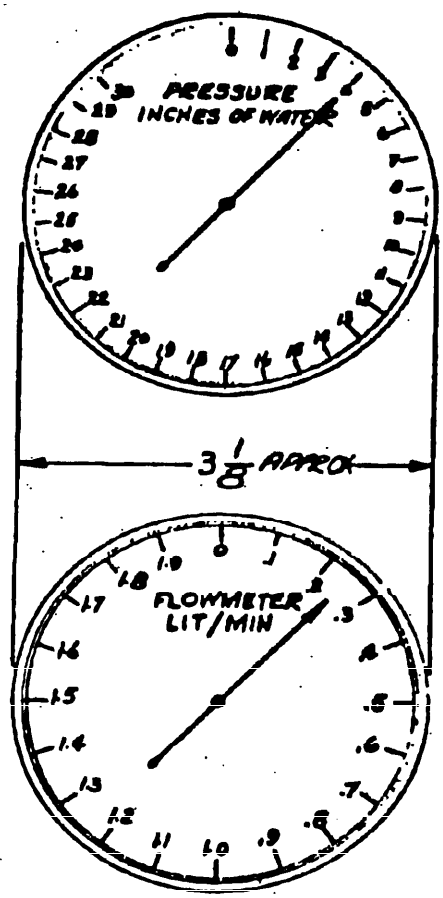


FIGURE 2. Tester, leakage, demand oxygen regulator, type MH-2
(dimensions in inches)

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3.7.5 Adapters. The adapters for connecting the tester to oxygen regulators shall be as shown in Figure 1. The stopper type shall be connected directly to the tester to regulator tubing and the second adapter shall be secured to the end of the tubing by means of a small, flexible chain conforming to RR-C-271, type II, class 6. The stopper shall be a commercial No. 2 rubber stopper. The other adapter shall conform to Figure 3.

3.7.6 Case. The case shall be fabricated from 0.062 inch aluminum conforming to QQ-A-250/8F, temper H34. The top shall be hinged to the body of the case. The cover shall be provided with a suitable latch for holding the cover closed when the tester is not in use. A handle for carrying the case shall be provided on the top of the case.

3.7.6.1 Finish. The case shall be given one coat of zinc-chromate primer, conforming to TT-P-1757 prior to painting. After the primer is thoroughly dry, the case shall be painted with two coats of lacquer conforming to TT-L-32. The color of the lacquer shall conform to No. 17038 of FED-STD-595.

3.8 Weight. The weight of the complete tester shall not exceed 6 pounds.

3.9 Identification of product. The tester shall be marked in accordance with MIL-STD-130.

3.10 Operating instructions. The operating instructions shall be securely attached to the inside of the case cover, positioned as shown in Figure 1, and placed on an easily replaceable card. The instructions shall be as follows:

1. Plug hose with connector from MH-2 tester into female connection of regulator hose, and close relief valve on bulb.

2. Test of A-12, A-12A, A-13, A-14, A-15, A-21, D-1, D-2, D-2A, MB-2, MD-1, or MD-2 regulators:

   (a) Squeeze bulb until pressure gage reads 17 inches of water and squeeze bladder to hold pressure.

   (b) Leakage flow gage should not read more than 1.0 liter per minute on "normal" nor more than 0.5 liter per minute on "100 percent."

3. If leakage is excessive, repeat test with tester, plugged into regulator outlet. If leakage is still excessive, regulator is defective; if it is no longer excessive, hose is defective.

4. If leakage exceeds the maximum allowable leakage, remove regulator for repair. Rework tag should read, "Removed for excessive outward leakage."

5. To check the MH-2 tester, close end of stopper with finger and squeeze bulb until pressure gage reaches 17 inches of water pressure. A reading of zero on the leakage flowmeter indicates the tester is operating satisfactorily.

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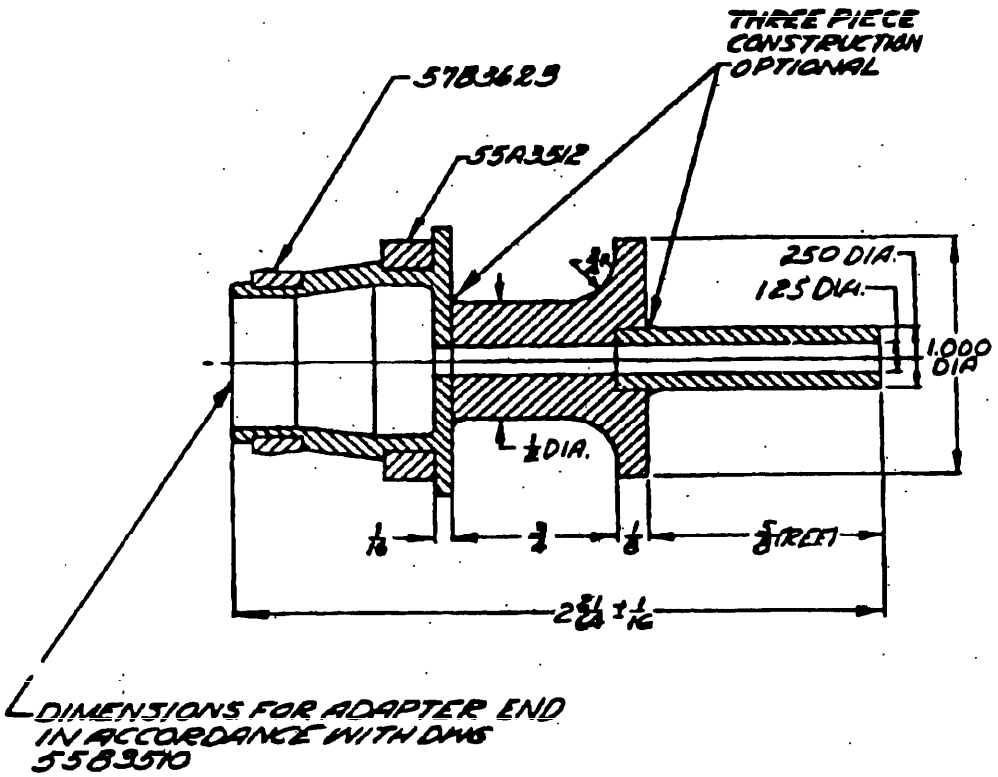


FIGURE 3. Adapter

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6. If it is impossible to reach 17 inches of gage pressure and according to test 5, above, tester is operating satisfactorily, the regulator is leaking excessively and should be removed for repair.

7. To test regulator technical order compliance with pressure breathing outlet pressures, read outlet pressure in inches of water on tester's pressure gage while the regulator is set as specified in the applicable technical order.

3.11 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.12 Workmanship. The tester shall be constructed in accordance with high-grade aircraft instrument manufacturing practice. The tester shall be thoroughly cleaned of all oil and foreign matter and free from any defects that might affect its operations or functionability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of testers shall be classified as follows:

a. Acceptance tests (see 4.2).

b. Preproduction testing (see 4.3).

4.2 Acceptance tests. Acceptance test shall consist of:

a. Individual tests.

b. Sampling tests.

4.2.1 Individual tests. Each tester shall be subjected to the following tests as described under Test Methods of this specification:

a. Examination of product.

b. Flow measuring gage.

c. Pressure gage.

d. Leakage.

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4.2.2 Sampling plan and tests.

4.2.2.1 Sampling plan. One tester shall be selected at random from each lot of 100 testers, or fraction thereof, and, subjected to the following tests:

a. Overload.

b. High and low temperature operation.

c. Shock.

4.2.2.1.1 Rejection. When one tester from a lot fails the specified tests, this is cause for rejection of the lot. Before resubmitting the lot, the contractor shall fully explain to the inspector both the corrections made and the cause of previous rejections.

4.3 Preproduction testing.

4.3.1 Preproduction test sample tested by the contractor. The contractor shall subject one tester to the preproduction tests specified in 4.3.4.

4.3.2 Preproduction test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of this report to the Government activity specified in 6.3.

4.3.3 Preproduction test sample for the Government procuring activity. Along with the preproduction test report, the contractor shall submit one sample to the Government activity specified in 6.3.

4.3.4 Preproduction tests. Preproduction tests shall consist of all tests described under test methods.

4.4 Test conditions.

4.4.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees Fahrenheit, plus or minus 18 degrees, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowances shall be made for the change in instrument reading.

4.5 Test methods.

4.5.1 Examination of product. Unless otherwise specified by the procuring activity, each tester shall be checked to determine compliance with this specification with respect to materials, workmanship, finish, weight, and markings.

4.5.2 Flow measuring gage. The calibration of the gage shall be tested with orifice attached end at conditions of 4.4.1. A downstream pressure of 17 inches of water shall be applied and flow adjusted to 0, 0.25, 0.50, 1.0, and 2.0 liter per minute as measured with an adjusted flow meter. Any gage indicating flows which vary more than the rate of plus or minus 1 percent based on full scale shall be rejected.

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4.5.3 Pressure gage. The pressure gage shall be attached to a water manometer having a range from 0 to 30 inches of water. The gage shall be tested at 5, 10, 20, and 25 inches of water pressure. Any gage indicating errors more than plus or minus three percent at any test points shall be rejected.

4.5.4 Leakage. The rate of outward leakage through the tester shall be determined by applying 20 inches of water pressure internally to the tester, with the regulator adapter plugged. The rate of leakage shall be determined by the pressure drop within the system. Testers indicating a leakage in excess of 0.50 inch of water pressure drop per minute or a deviation of the flow measuring gage from zero (in accordance with Section 5 of 3.10, operating instructions), shall be rejected.

4.5.5 Overload. A mercury manometer with a range of at least 0 to 30 inches of mercury shall be attached to the regulator adapter with the adapter plugged. The internal pressure shall be increased by means of the pressure bulb until a pressure of 3 inches of mercury is indicated on the manometer. This pressure shall be maintained for a period of 2 minutes. The flow measuring gage and pressure gage shall indicate within the allowable error specified herein at the completion of this test when subjected to the individual tests.

4.5.6 High and low temperature operation. The pressure gage and the differential flow gage shall be subjected to a temperature of plus 160 degrees Fahrenheit and minus 65 degrees Fahrenheit, respectively, for a period of 2 hours. A means of supplying a known static half-scale reading shall be provided to each gage; any gage deviation of more than the rate of plus or minus one percent based on full scale shall be rejected. The completed and calibrated tester (as per 4.5.2) shall continue to operate satisfactorily after being subjected to plus 160 degrees Fahrenheit and to minus 65 degrees Fahrenheit for 2 hours. No flow measuring correction is necessary for requirements of tester.

4.5.7 Shock. One complete tester shall be tested by dropping it on a concrete slab from a height of five feet. A total of 9 blows shall be applied, three paralleled to each of the three principal axes of the tester in order of back, top, and side. At the completion of the shock test, the gages shall be checked for calibration errors and examined for any evidence of damage to the internal parts. Any deviation from requirements of 3.6.8 shall be cause for rejection. This is considered a destructive test. Samples tested for shock shall not be submitted as part of the production order. Upon completion of this test, the tester shall be subjected to and meet 4.5.4.

* 5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be level A, C or industrial in accordance with MIL-C-52211, as specified (see 6.2).

* 5.2 Packing. Packing shall be level A, B, C or industrial in accordance with MIL-C-52211, as specified (see 6.2).

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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5.3.1 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. This tester is intended for use in testing demand type oxygen regulators for outward leakage while installed in the aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Preproduction requirements.

b. Level of packing and packaging required.

c. Unit quantities.

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Air Force - 99

Preparing activity:
Air Force - 82

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